

Guía de Generación de Documentación

En este proyecto, hay dos componentes de la documentación que son generados por código, y por lo tanto pueden ser fácilmente actualizados para estar a la par del código. Este documento describe su utilidad y como generarse.

Referencia del código

La referencia de código es una serie de archivos documentados en el archivo web [“Documentation Reference.html”](#). Este archivo se genera a través de la ejecución del programa de Python `documentation_creator.py`

Ejemplo de ejecución:

```
Anaconda Powershell Prompt (Anaconda3)
(base) PS C:\Users\Fedacking\source\piton\Microscopy-Program> python .\documentation_creator.py
Socket: connect operation failed with error 10061
wrote Backend.camera_backend.html
wrote Backend.experiment_backend.html
wrote Backend.focus_backend.html
wrote Backend.frequency_backend.html
wrote Backend.lines_backend.html
wrote Backend.lockin_backend.html
wrote Backend.main_backend.html
wrote Backend.platina_backend.html
wrote Backend.points_backend.html
wrote Backend.rect_backend.html
wrote Drivers.Camera.virtual_camera.html
wrote Drivers.camera_selector.html
wrote Drivers.DAQ.daq_driver.html
wrote Drivers.DAQ.mc_usb_2527_driver.html
wrote Drivers.DAQ.virtual_backend.html
wrote Drivers.daq_selector.html
wrote Drivers.FunGen.sdg5122_driver.html
wrote Drivers.FunGen.virtual_fungen.html
wrote Drivers.fungen_selector.html
wrote Drivers.Lockin.anfatec_driver.html
wrote Drivers.Lockin.LI5655.html
wrote Drivers.lockin_selector.html
wrote Drivers.Motor.MotorDriver.html
wrote Drivers.motor_selector.html
wrote Model.frequency_scan.html
wrote Model.operation.html
```

En una subcarpeta crea todos los archivos HTML que corresponden 1 a 1 con los módulos de Python.

Ejemplo de como se ve la referencia:

Documentation Reference

»	
Name	
	Backend.camera_backend.html
	Backend.experiment_backend.html
	Backend.focus_backend.html
	Backend.frequency_backend.html
	Backend.lines_backend.html
	Backend.lockin_backend.html
	Backend.main_backend.html
	Backend.platina_backend.html

Model.point

Classes

[builtins.object](#)
[Point](#)

class **Point**([builtins.object](#))

[Point](#)(x: float, y: float, freq: int, n=1, display_x: int = 0, display_y: int = 0)

Class that represents a point to be measured in the round

Methods defined here:

```
__init__(self, x: float, y: float, freq: int, n=1, display_x: int = 0, display_y: int = 0)
    :param x: X coordinate in the camera that represented this point
    :param y: Y coordinate in the camera that represented this point
    :param freq: Frequeency at wich the point should be measured
    :param n: Amount of times the point should be re read to average the results
    :param display_x: X coordinate where the point should be represented in a map of the operation
    :param display_y: Y coordinate where the point should be represented in a map of the operation
```

str (self)

El texto de la clase en el formato pydoc se utiliza para generar el código. Acá podemos ver el código de la clase Points que se uso para generar este archivo de documentación:

```
class Point:
    """Class that represents a point to be measured in the round"""

    def __init__(self, x: float, y: float, freq: int, n=1, display_x: int = 0, display_y: int = 0):
        """
        :param x: X coordinate in the camera that represented this point
        :param y: Y coordinate in the camera that represented this point
        :param freq: Frequeency at wich the point should be measured
        :param n: Amount of times the point should be re read to average the results
        :param display_x: X coordinate where the point should be represented in a map of the operation
        :param display_y: Y coordinate where the point should be represented in a map of the operation
        """
```

Diagrama UML

Usando una consola en la carpeta principal se puede usar esto para crear la imagen.

pyreverse -o png -p Microscopy -Smy View\program_window.py

```
(base) PS C:\Users\Fedacking\source\python\Microscopy-Program> pyreverse -o png -p Microscopy -Smy View\program_window.py
parsing View\program_window.py...
```

Este archivo tiene todas las clases y sus relaciones de pertenencia. Esto permite ver la relación del código.

